Slide 1:

Introduction:

Good Moring to all of you,

My self ALLAM SRIDHAR, am here with my team members are here to present our Dashboard/Analysis.

This ppt was created to provide Excelerate with a comprehensive view of their user engagement and opportunity completion across various demographics, locations, and skills. The data reflects insights into the platform's user base and program effectiveness, helping Excelerate understand both who is using the platform and how they are engaging with it.

Slide 2:

Overview of Todays Presentation

Slide 3:

Preprocessing:

- In SLU Opportunitywise dataset there are around 33 features and 8556 records
- It contains information on opportunity signups, personal details, opportunity start and completion dates,reward amount and badges earned, etc…
- There are some missing values(1-5) in City, state, zipcode, institution name, current/intended major columns and missing most of the values(3794-8529) in Address Line 2, Opportunity Start Date, Reward Awarded Date, Completion Date, Reward Amount, Badge Id, Badge Name, Skill points Earned, skill Points.

Missing values handling:

- Here i dont think Address Line 1, Address line 2 will be of much use, so i think of replacing them with Null/Not mentioned.
- There is 1 missing in city, 3 in state,3 in zip codes, first ill try to find their values through adjacent city, state, city/address columns if found then okay else ill remove them directly , since they are very few in numbers.
- There are 5 missing in Institution name and Current/Intended Major/Institution Name, most probably ill try to group by country and then take most frequent in that country and replace with them.
- There are 3794 null values in the Opportunity start date, here since there is Opportunity End Date, ill try to find if there is any relation between those columns and replace the null values with those relationship values(i.e days difference).
- There are 8529 missing values in Reward Awarded Date, Completion Date, Reward Amount, Badge Id, Badge Name, Skills Points Earned, Skills Earned columns, actually these values are present only if the student has completed his course/opportunity, so since these values are

missing indicates that, the remaining students have not completed the course/opportunity, so i like to replace with 0 for numerical cols and Null for categorical columns.

Standardizing columns/values:

- In country,state, city Institution Name column for the same place they have entered the values differently like Iran Islamic Republic of Persian Gulf and Iran, Islamic Republic of Persian Gulf both are same so replacing them with Iran only using the mapping dictionary. Similarly we created and used mapping dictionaries for state, city, Institution Name columns to solve this problem
- We also standardized the zip column like extracted only numeric value, also removed the spaces in between numbers, also removed any special char like hyphen(-) in between etc…
- Also Converted the Date columns from object to DateTime and extracted only date from it

Feature Engineering:

- Adding Address line 1 and Address line 2 together and call it as Full Address
- Adding First Name and Second Name together and calling it as Full Name
- Extracting Age from Date of Birth Column
- Extracting Engagement Duration to know how many days he has been engaged on this platform
- Dropping the First Name, Last Name, Address Line 1, Address Line 2 columns since we have created new features

Slide 4:

EDA

Univariate analysis

- From here we can see that Most of the Age of participants is in between around 22-26 years
- In reward Amount there are only 2 distinct values 50 and 200, i.e. after completeion of opportunity the reward amount which candidates received are either 50 or 200 only so far.
- In status Description we can see that most of the opportunities applied got rejected
- And most of the people who are applying are applying for internships and few for course.
- Among all the countries the majority of the participants are from United states followed by India.
- In state wise distribution, I can clearly see that most of the students are from Missouri, followed by Telangana and Andhra Pradesh.
- In city wise distribution I can clearly see that the majority of the participants are from Saint Louis City.
- In the bottom left we can see the top 10 Opportunity names, among which Career Essentials, Getting started with your professional journey course tops the list, followed by data visualization.
- Among top 10 majors I can clearly see that most of those are from Computer science background Majors.

- Here in the badge names, these are the badges which were awarded to participants who had completed some particular opportunity.
- In the last chart we can see the top 10 people names who have applied for more number of opportunities so far.

Slide 5:

Bivariate:

- In first chart, we can see that in all the opportunity Categories there is more male participation than the other Genders.
- In the second chart we can see the Gender distribution across top 10 countries, over here we can see that there all these countries are male dominant countries, so I tried to find if there are any countries when the female participation is more than male participation, whose result we can see in chart 4
- In the current student statues by current/Intendent Major(top 10) chart, we can see that exept for Computer science and engineering and Computer engineering where under-graduating participation is more, in all other Major are dominated by Graduating students

Slide 6:

Now let us look at the Completion Rate

- Overall if we see, there is only 29 records out of 8558 records which can be considered Completed, so the overall completion rate is 0.34% only.
- In first chart i.e. Top 10 opportunities by completion rate, here we can only see 4 opportunities cause till now only some opportunitites from these four have been completed so far.
- In second chart we can see that the completion rate of female is slightly higher than that of male.
- In third chart we can see that completion rate of Gratduate students is 0.4 and of undergraduate students is 0.056
- When it comes to completion rates across countries, United States has the higher completion rate (0.529), followed by Ghana(0.364), followed by India(0.211) and followed by Nigeria (0.132)
- In the next chart these are the candidates who have successfully completed more opportunities so far.
- In the last chart we can see that for Opportunities like CPR/AED Certification, Carrer Essentials: Getting Started with Your Professional Journey the Rejections are very less compared to all other opportunities.

Slide 7:

Trend Analysis:

- **In Number of signups per year** chart, Here, the data is of range from January 2023 to march 2024.

- Even though the Entire Signups in 2023 are approximately around 5700, but In just 3 months of 2024
- The signups are around 2800,indicating that there is an Upward trend in Signups over the years.
- **In Number of applications by year** chart, We can see that over the years the number of applications have been increasing enormously, also the 2024 has more Applications, but this 4801 applications in this year are from the first 3 months of the year only, actually we can estimate applications total of 3 times of this value by the end of this year.
- **In Graduation Trend by year chart**, Compared to Graduated there are more graduating students, who are expected to be graduated in next 1-4 years.
- **In opportunity Start trend chart**, Here we can see that in 2022 year there were more than 5500 opportunities started in this year, but there is sudden downfall in number of opportunities starting in 2023. But in 2024 we can again see that the number of opportunities starting are more
- **In opportunity End trend chart**, Here we can see that most of the opportunities are being ended in this year i.e. 2024
- In monthly completion rate chart, Here we can see that most of the opportunities are being completed in the months of October, then some of them have been completing in September and February months

Slide 8:

In Monthly Trends we can see that

- There are huge spikes in the months of June to august when it comes to signups.
- When it comes to application trend here we can see huge spikes in the months of January to March 2024 indicating high applications.
- Here we can also see that more number of opportunities are being started in the months of October.
- More number of opportunities are ending in the months of February to April.

Slide 9:

Churn Analysis and Predictive Modeling:

In churn analysis, we utilise advanced machine learning techniques to predict and understand the factors leading to student drop-offs. The primary goal is to identify at-risk students early, allowing for timely and targeted interventions to improve retention rates. Understanding churn is critical in educational settings, as reducing drop-offs not only enhances student outcomes but also contributes to the sustainability and success of academic programs.

- Here Data Cleaning and Feature Engineering steps are already covered in preprocessing, Here we also Encoded categorical Features, created a churn Label if value is 1 drop out and if 0 not drop out, then also we have created few derived features such ass Processing time, etc…
- In model building we did use build the model using different algorithms whose results let us see in next slide.

- Then we also calculated metrices for Model Evaluation and then also found the feature importance.

Slide 10:

Now here these are the various models and metrices we used to build models so far, out of which we can see that Logistic Regression, Support vector Machine and K-Nearest Neighbour are not suitable, about remaining with some more tuning we can use them to increase the performance

Slide 11:

Recommendation Systems

Here we did create 3 different recommendation Systems

1.Course recommendation system

2.Risk assessment recommendation system

3.Recomendation system for individual candidates , dropout probability, risk factor, dropout-factor and recommendation for that factor.

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Course Recommendation system: if we pass the student details to the model it returns the recommended course for that student and gives some Engagement Recommendations (i.e. it also gives recommendations like how to get engaged for not dropping out). Here one example result for test case you can see on this slide

Slide 13:

In this Risk assessment Recommendation system also if we pass the student details then this model returns Personalized recommendations, tells the risk level and Risk suggested Actions for that candidate, one example test case is given on this slide

Slide 14:

In this recommendation system if I pass the student details to the model, it gives top 5 risk factors, Risk probability, Risk level, Risk factors and recommendation for that specific factor, also some Recommended Interventions too.